

**FORLENS – A LIGHTWEIGHT AI-DRIVEN ENDPOINT SECURITY SOLUTION FOR
REAL-TIME THREAT DETECTION IN SMES**

25-26J-076

Thesis Group Report

Nusfa M R F – IT22908742

Sanjula E A Y – IT22340078

Salgado M. B. D – IT21289934

De Silva H S- IT22887580

Bsc (Hons) in Information Technology Specializing in Cyber Security

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

September 2025

**FORLENS – A LIGHTWEIGHT AI-DRIVEN ENDPOINT SECURITY SOLUTION FOR
REAL-TIME THREAT DETECTION IN SMES**

25-26J-076

Thesis Group Report

Nusfa M R F – IT22908742

Sanjula E A Y – IT22340078

Salgado M. B. D – IT21289934

De Silva H S- IT22887580

Bsc (Hons) in Information Technology Specializing in Cyber Security

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

September 2025

DECLARATION

ABSTRACT

The escalating complexity of the global cyber threat landscape has precipitated a widening security gap, particularly affecting Small and Medium Enterprises (SMEs) that constitute over 90% of global businesses. Small and Medium Enterprises (SMEs) are increasingly exposed to sophisticated cybersecurity threats, including insider attacks, lateral movement, data exfiltration, and advanced persistent threats. However, due to limited financial resources and technical expertise, SMEs often lack access to enterprise-grade security solutions such as Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Security Orchestration, Automation, and Response (SOAR) platforms. This creates a significant security gap, making SME environments highly vulnerable to both internal and external threats.

This dissertation presents **ForLENS**, a lightweight, AI-driven endpoint security framework designed to provide real-time threat detection, automated response, and tamper-proof forensic logging tailored for SME environments. The proposed system integrates four core components: a resilient endpoint monitoring agent with embedded SMB-based deception capabilities, a machine learning-driven anomaly detection and threat intelligence engine, a blockchain-based secure logging mechanism ensuring data integrity and non-repudiation, and a SOAR-inspired alert interface for visualization and automated response. The methodology involves continuous endpoint telemetry collection, behavioral anomaly detection, enrichment using open-source threat intelligence feeds, cryptographic log hashing with blockchain anchoring, and automated incident response mechanisms.

The system is evaluated using simulated attack scenarios, including unauthorized access attempts, abnormal user behavior, malicious file execution, and insider threat activities. Experimental results demonstrate real-time detection with low resource overhead, efficient automated containment, and reliable tamper detection of security logs. The system also provides explainable alerts, improving usability for non-expert users.

In conclusion, ForLENS offers a cost-effective, scalable, and integrated cybersecurity solution that combines detection, response, and forensic capabilities within a lightweight architecture. The system addresses critical limitations of existing solutions and enhances cybersecurity

resilience in SME environments. Future improvements may include advanced machine learning integration and large-scale real-world deployment.

Keywords

Cybersecurity, Endpoint Security, Insider Threat Detection, Machine Learning, Threat Intelligence

ACKNOWLEDGEMENTS

The successful completion of this group dissertation would not have been possible without the extensive assistance and professional guidance provided by the faculty of the Sri Lanka Institute of Information Technology (SLIIT). Sincere appreciation is extended to our supervisor, Mr. Kanishka Yapa, and co-supervisor, Mr. Harinda Fernando, for their technical insights into cryptography, endpoint surveillance, and secure log integration.[1, 1] We are also grateful to the SMEs that participated in our initial pilot discussions, providing real-world context into the challenges of resource-constrained cybersecurity. Finally, acknowledgement is due to the developers of the open-source tools—including Ethereum Geth, Python psutil, and various lightweight SOAR frameworks—which served as the foundational components of the ForLens ecosystem.

TABLE OF CONTENTS

DECLARATION	3
ABSTRACT	4
ACKNOWLEDGEMENTS	6
TABLE OF CONTENTS.....	7
LIST OF TABLES	8
1. INTRODUCTION.....	9
1.1. Background Literature Survey.....	9
1.2. Research Gap Identification.....	12
1.3. Research Problem and Objectives	15
2. Methodology.....	18
2.1. over all methodology	18
2.2. System Architecture / Framework Design	21
2.3. Description of Components (All Four Modules)	23
2.4. Data Flow & Integration.....	26
2.5. Tools and Technologies Used.....	28
2.6. Commercialization Aspect of the Product	31
2.7. Testing and Implementation.....	34
3. RESULTS AND DISCUSSION.....	37
3.1. Results.....	37
3.2. Research Findings	38
3.2.1. Deception Efficacy and Operational Resilience.....	39
3.2.2. Storage Efficiency vs. Verification Speed	39
3.3. Discussion.....	39
3.4. Summary of Each Student's Contribution.....	40
4. Conclusion.....	41
4.1. Future Work	42
5. References	43

LIST OF TABLES

Table 1: Security Approach Comparison 12
Table 2: Test Cases 38

1. INTRODUCTION

1.1. Background Literature Survey

The rapid advancement of digital technologies, cloud computing, and interconnected network infrastructures has significantly transformed the operational landscape of modern organizations, leading to an increased reliance on digital systems for storing, processing, and transmitting critical information. This digital transformation has enabled organizations to achieve higher levels of efficiency, scalability, and global connectivity. However, it has simultaneously introduced new security challenges by expanding the attack surface, making cybersecurity a fundamental concern across all sectors. In particular, Small and Medium Enterprises (SMEs) have emerged as highly vulnerable targets due to their limited financial resources, lack of specialized cybersecurity expertise, and inadequate implementation of advanced security mechanisms. Studies indicate that SMEs often struggle to adopt robust security frameworks despite being increasingly targeted by cybercriminals, resulting in significant economic and operational consequences [9], [15], [31], [32].

Unlike large enterprises that deploy comprehensive and resource-intensive security solutions such as Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Security Orchestration, Automation, and Response (SOAR) systems, SMEs typically rely on basic security tools that focus primarily on signature-based detection. While these traditional approaches are effective against known threats, they are inadequate for detecting modern attack vectors such as insider threats, lateral movement, zero-day exploits, and advanced persistent threats (APTs). Industry reports have consistently highlighted that cyber incidents continue to increase in both frequency and complexity, leading to severe financial losses and reputational damage for organizations that lack advanced defensive capabilities [11], [16], [17]. Furthermore, insider threats pose a particularly challenging problem, as they originate from trusted users within the system, making them difficult to detect using conventional methods [12], [18], [33]. This limitation emphasizes the need for more intelligent and adaptive security mechanisms that can detect anomalies beyond predefined signatures.

In response to these limitations, recent research has focused on the application of machine learning and artificial intelligence techniques for anomaly-based threat detection. Machine

learning models such as Isolation Forests, Autoencoders, and deep learning-based approaches have been widely adopted to analyze behavioral patterns and identify deviations that may indicate malicious activities [22], [23], [25]. These models establish baseline behavior profiles and continuously monitor system activities to detect anomalies in real time, enabling the identification of previously unknown threats. Techniques such as DeepLog further enhance this capability by analyzing system logs to detect abnormal patterns and predict potential system failures or attacks [24]. Additionally, explainable artificial intelligence techniques, including SHAP and LIME, have been introduced to improve the interpretability of machine learning models, allowing security analysts to understand the reasoning behind anomaly detection decisions and thereby increasing trust in automated systems [26], [30].

Another important advancement in modern cybersecurity is the integration of threat intelligence platforms, which provide real-time information about known malicious indicators such as IP addresses, domains, and file hashes. Standards such as Structured Threat Information Expression (STIX) and Trusted Automated Exchange of Intelligence Information (TAXII), along with platforms like Malware Information Sharing Platform (MISP), enable organizations to share and consume threat intelligence effectively [27], [28], [29]. The use of open-source threat intelligence feeds further enhances detection accuracy by providing up-to-date information on emerging threats, allowing systems to correlate observed activities with known attack patterns [13], [14]. This integration significantly improves situational awareness and reduces false positives, particularly in dynamic threat environments where attack patterns evolve rapidly.

The integrity and reliability of security logs have also become a critical aspect of cybersecurity, especially in the context of incident response and digital forensics. Traditional logging mechanisms are vulnerable to tampering, enabling attackers to modify or delete evidence to conceal their activities. To address this issue, cryptographic techniques such as forward-secure logging and hash chaining have been proposed to ensure data integrity and non-repudiation [35], [36], [37]. More advanced approaches incorporate blockchain technology to provide immutable and tamper-proof storage of log data, ensuring that once recorded, logs cannot be altered or deleted [38], [39], [41]. These mechanisms are further strengthened by the use of cryptographic structures such as Merkle trees, which enable efficient verification of data integrity [40]. Such secure logging systems play a crucial role in forensic investigations and compliance with

regulatory standards such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), which mandate secure handling and auditing of sensitive information [42], [43].

In addition to detection and logging mechanisms, deception-based cybersecurity techniques have gained significant attention as proactive defense strategies. Honeypots and decoy systems are designed to simulate vulnerable resources within a network, attracting attackers and enabling early detection of malicious activities. Research has demonstrated that honeypots can effectively capture attacker behavior and provide valuable insights into attack strategies, thereby improving threat detection and response capabilities [1], [2], [8]. High-interaction and adaptive honeypots further enhance this capability by dynamically responding to attacker behavior and generating realistic attack environments [6], [7]. The integration of machine learning with honeypot systems has also been explored to improve their adaptability and effectiveness in detecting sophisticated attacks [6]. These techniques are particularly useful for identifying lateral movement and insider threats, which are often difficult to detect using traditional monitoring systems.

Despite these advancements, existing cybersecurity solutions remain fragmented, complex, and resource-intensive, limiting their applicability in SME environments. Many systems require significant computational resources, continuous monitoring, and skilled personnel to operate effectively, creating barriers to adoption for smaller organizations. Furthermore, the lack of integration between different security components, such as monitoring, detection, logging, and response systems, results in inefficiencies and delays in threat mitigation. Research has shown that integrated and automated solutions, such as SOAR platforms, can significantly improve incident response efficiency by reducing manual intervention and enabling faster decision-making [3], [4], [45]. However, these solutions are often designed for large enterprises and may not be suitable for SMEs due to their complexity and cost.

Therefore, there is a critical need for a lightweight, cost-effective, and integrated cybersecurity framework that addresses the specific requirements of SMEs. Such a framework should combine real-time endpoint monitoring, intelligent anomaly detection, threat intelligence integration, secure and tamper-proof logging, and automated response capabilities within a unified architecture. By integrating these components, it is possible to enhance overall security posture,

improve detection accuracy, and reduce response time while maintaining low resource consumption. The development of such integrated solutions represents a significant step toward bridging the gap between advanced cybersecurity technologies and their practical deployment in resource-constrained environments, ultimately enabling SMEs to achieve a higher level of resilience against modern cyber threats.

Security Approach	Core Mechanism	SME Suitability	Limitations
Traditional SIEM	Rule-based aggregation	Low	High cost, alert fatigue
Enterprise SOAR	Manual playbooks	Low	Requires skilled analysts, high TCO
Blockchain (Naive)	Raw data on-chain	Low	Massive storage overhead
Honey-Agent (ForLens)	Embedded host-level deception	High	Prototype focused on SMB lures
ForLens Resilience	Watchdog self-recovery	High	Minimal impact on user productivity

Table 1: Security Approach Comparison

1.2. Research Gap Identification

Despite the rapid advancements in cybersecurity technologies and the growing body of research addressing various aspects of cyber defense, a significant gap remains in the development of integrated, lightweight, and cost-effective security solutions specifically tailored for Small and Medium Enterprises (SMEs). Existing research has largely focused on individual components of cybersecurity, such as anomaly detection using machine learning, threat intelligence integration, secure logging mechanisms, and automated response systems. While each of these approaches has demonstrated effectiveness in isolation, they are often implemented as standalone solutions without seamless integration into a unified framework. This fragmented approach requires organizations to deploy multiple independent tools, leading to increased operational complexity, higher implementation and maintenance costs, and inefficiencies in threat detection and response processes [20], [31], [32]. Such challenges are particularly pronounced in SME environments,

where limited resources and technical expertise make it difficult to manage and coordinate multiple security systems effectively.

Furthermore, most advanced cybersecurity solutions are designed with large-scale enterprise environments in mind, assuming the availability of high computational resources, continuous cloud connectivity, and specialized cybersecurity personnel. These assumptions do not align with the practical constraints of SMEs, which often operate with minimal infrastructure and lack dedicated security teams. Reports and studies have highlighted that SMEs are disproportionately affected by cyber threats while simultaneously lacking the capability to deploy enterprise-grade solutions such as SIEM, EDR, and SOAR platforms due to their complexity and cost [15], [17], [21]. As a result, SMEs frequently rely on basic security mechanisms that are insufficient to address modern threat landscapes, thereby creating a critical vulnerability gap that attackers can exploit.

Another major limitation in existing systems is the continued reliance on signature-based detection techniques in many endpoint security solutions. Although such methods are effective against known threats, they are incapable of identifying sophisticated attacks such as insider threats, lateral movement, and zero-day exploits. Insider threats, in particular, remain a persistent challenge due to their inherent complexity, as they involve legitimate users performing malicious actions that are difficult to distinguish from normal behavior [12], [18], [33]. To address this issue, machine learning-based anomaly detection techniques have been proposed, leveraging models such as Isolation Forests, Autoencoders, and deep learning architectures to identify deviations from normal behavioral patterns [22], [23], [25]. However, despite their effectiveness, many of these models lack explainability, making it difficult for non-expert users to interpret the results and trust automated decision-making processes. The absence of explainable outputs limits the practical usability of such systems, especially in SME environments where users may not possess advanced technical knowledge [26], [30].

In addition to detection challenges, the integrity and reliability of security logs remain a critical concern in modern cybersecurity systems. Many existing solutions fail to implement robust and tamper-proof logging mechanisms, leaving them vulnerable to log manipulation or deletion by attackers after a system compromise. This limitation significantly undermines the effectiveness

of forensic investigations and incident response processes. Although cryptographic approaches such as forward-secure logging, hash chaining, and blockchain-based logging have been proposed to address these issues, their integration into lightweight and resource-efficient systems remains limited [35], [36], [38], [39]. As a result, there is a lack of practical solutions that can provide both strong security guarantees and operational feasibility for SMEs.

Moreover, deception-based techniques such as honeypots have been extensively studied for their ability to proactively detect attacker behavior by simulating vulnerable resources within a network environment. These techniques have proven effective in capturing attack patterns and identifying malicious activities at early stages [1], [2], [8]. However, most existing honeypot implementations are deployed as separate infrastructure components rather than being integrated directly within endpoint systems. This separation limits their effectiveness in detecting real-time threats at the host level, particularly in SME environments where centralized infrastructure may be limited. The integration of lightweight, adaptive honeypot mechanisms within endpoint agents remains an underexplored area, presenting an opportunity for innovation in cybersecurity system design [6], [7].

Additionally, while recent advancements have introduced automation and orchestration through SOAR platforms to improve incident response efficiency, these solutions are often complex and require significant configuration and expertise to operate effectively. Although they offer substantial benefits in reducing response time and automating repetitive tasks, their adoption in SME environments is limited due to high implementation costs and operational complexity [3], [4], [45]. This further emphasizes the need for simplified and user-friendly frameworks that can deliver similar capabilities without imposing excessive resource or knowledge requirements.

Overall, the existing body of research highlights a clear lack of unified cybersecurity frameworks that integrate multiple critical components—such as real-time endpoint monitoring, intelligent anomaly detection, threat intelligence enrichment, secure and tamper-proof logging, and automated response—into a single cohesive system. Most solutions address these aspects independently, resulting in fragmented security architectures that are difficult to manage and inefficient in practice. Therefore, there is a significant research gap in the development of an integrated, lightweight, and scalable cybersecurity solution that is specifically designed to meet

the constraints and requirements of SMEs. Addressing this gap requires a holistic approach that combines advanced detection techniques, secure logging mechanisms, and automated response capabilities within a unified architecture, while ensuring usability, efficiency, and cost-effectiveness. Such a solution has the potential to significantly enhance the cybersecurity posture of SMEs by providing real-time threat detection, explainable insights, reliable forensic support, and efficient incident response within a practical and deployable framework

1.3. Research Problem and Objectives

Small and Medium Enterprises (SMEs) face a growing challenge in protecting their digital assets against increasingly sophisticated cyber threats due to inherent limitations in financial resources, technical expertise, and access to advanced cybersecurity infrastructure. As organizations continue to digitize their operations and rely heavily on interconnected systems, they become more exposed to a wide range of cyber risks, including insider threats, advanced persistent threats, lateral movement, and zero-day exploits. Despite this increased exposure, SMEs often lack the capability to implement comprehensive security solutions, making them highly attractive targets for attackers. Reports and studies have consistently highlighted that SMEs are disproportionately affected by cyber incidents, suffering significant financial losses and operational disruptions due to inadequate security measures [9], [15], [16].

Existing cybersecurity solutions, such as Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Security Orchestration, Automation, and Response (SOAR) platforms, provide advanced capabilities for threat detection, monitoring, and response. However, these systems are typically designed for large-scale enterprise environments and require substantial computational resources, continuous monitoring, and skilled personnel to deploy and manage effectively [20], [21], [45]. As a result, they are often too complex and costly for SMEs to adopt. On the other hand, simpler security tools commonly used in SME environments rely primarily on signature-based detection techniques, which are ineffective against modern and evolving threats. These tools are unable to detect sophisticated attacks such as insider threats and abnormal user behavior, which require behavioral analysis and anomaly detection mechanisms [12], [18], [33].

Another significant limitation of existing systems is the lack of integration across different security functions. Many solutions operate independently, focusing on specific tasks such as monitoring, detection, logging, or response, without providing a unified approach. This fragmented architecture leads to inefficiencies in threat management, delayed incident response, and increased operational complexity. The absence of real-time, behavior-based detection mechanisms further exacerbates this issue, as organizations are unable to identify unknown or emerging threats in a timely manner. Although machine learning-based approaches have been introduced to address this limitation, many of these models lack explainability, making it difficult for non-expert users to interpret detection results and trust automated decisions [26], [30].

In addition to detection and response challenges, the reliability of security logs remains a critical concern in existing cybersecurity systems. Traditional logging mechanisms are vulnerable to tampering, allowing attackers to alter or delete evidence after compromising a system. This compromises the integrity of forensic investigations and reduces the effectiveness of incident response processes. Although cryptographic techniques such as forward-secure logging and blockchain-based logging have been proposed to ensure immutability and tamper resistance, their adoption in lightweight and resource-constrained environments remains limited [35], [36], [38], [39]. Furthermore, many existing systems do not incorporate proactive detection mechanisms such as deception techniques, including honeypots, which have been shown to be effective in identifying attacker behavior and detecting unauthorized access attempts at early stages [1], [2], [8].

Therefore, the core research problem addressed in this study is the lack of an integrated, lightweight, and cost-effective cybersecurity framework that can effectively support SMEs in detecting, analyzing, and responding to modern cyber threats. Specifically, there is a need for a unified solution that combines continuous endpoint monitoring, intelligent anomaly detection, threat intelligence integration, secure and tamper-proof logging, and automated response capabilities within a single cohesive architecture. Such a framework should be designed to operate efficiently within resource-constrained environments while maintaining high detection accuracy and usability for non-expert users.

To address this research problem, the primary objective of this study is to design and develop a comprehensive cybersecurity framework that integrates multiple security functionalities into a unified system tailored for SME environments. The framework aims to provide real-time visibility into system activities through lightweight endpoint monitoring, detect anomalies using machine learning-based techniques, enhance detection accuracy through threat intelligence enrichment, ensure the integrity and reliability of security logs using cryptographic and blockchain-based mechanisms, and automate incident response to minimize human intervention and reduce response time. Additionally, the framework seeks to improve usability by incorporating explainable detection mechanisms that allow users to understand and interpret system outputs effectively. By achieving these objectives, the proposed solution aims to bridge the gap between advanced cybersecurity technologies and their practical implementation in SMEs, ultimately enhancing their resilience against modern cyber threats.

2. Methodology

2.1. over all methodology

The methodology of the proposed ForLENS system is designed to provide a comprehensive, integrated, and lightweight cybersecurity framework capable of delivering real-time threat detection, automated response, and secure forensic logging within resource-constrained Small and Medium Enterprise (SME) environments. The system adopts a modular and layered architectural approach, where each component is responsible for a specific function while collectively contributing to an end-to-end security pipeline. This methodology is centered around continuous endpoint monitoring, behavioral anomaly detection, threat intelligence enrichment, tamper-proof logging, and automated incident response, ensuring that the system is capable of detecting, analyzing, responding to, and securely recording cybersecurity events in real time. Such integrated approaches align with modern security architectures that emphasize automation, correlation, and unified threat management rather than isolated security mechanisms [20], [4].

The overall operation of the system begins at the endpoint level, where a lightweight monitoring agent is deployed to continuously observe system-level activities, including process execution, file system modifications, network communications, and user interactions. Endpoint-level visibility is a critical requirement in modern cybersecurity, as many attacks originate from compromised hosts and propagate internally within a network [21]. The agent is designed to operate in a stealthy and resource-efficient manner, ensuring minimal impact on system performance while maintaining persistent monitoring capabilities. In addition to passive monitoring, the agent integrates a deception-based mechanism in the form of an embedded Server Message Block (SMB) honeypot, which simulates vulnerable resources to attract potential attackers. Deception technologies such as honeypots have been widely recognized for their ability to capture attacker behavior and detect unauthorized access attempts at early stages [1], [2], [8]. This proactive approach enables the detection of lateral movement and insider threats, which are often difficult to identify using traditional monitoring techniques.

The collected telemetry data is then forwarded to the central analytical component of the system, where machine learning-based anomaly detection techniques are applied to identify deviations from established behavioral baselines. The system utilizes a combination of unsupervised and

semi-supervised learning models, enabling it to detect both known and unknown threats without relying solely on predefined signatures. Techniques such as Isolation Forests and deep learning-based anomaly detection models have proven effective in identifying abnormal system behavior in dynamic environments [22], [25]. Furthermore, log-based anomaly detection approaches, such as DeepLog, enhance the system's ability to analyze sequential system events and detect subtle anomalies that may indicate malicious activity [24]. To enhance detection accuracy and contextual awareness, the system integrates threat intelligence feeds obtained from external platforms. Standards such as STIX and TAXII, along with platforms like MISP, facilitate structured threat intelligence sharing and enable the system to correlate detected anomalies with known malicious indicators [27], [28], [29]. This combination of behavioral analysis and intelligence-driven enrichment enables the generation of high-confidence alerts while minimizing false positives.

Once a potential threat is identified, the system triggers automated response mechanisms inspired by Security Orchestration, Automation, and Response (SOAR) principles. These mechanisms enable the system to take immediate action, such as isolating compromised endpoints, terminating malicious processes, blocking suspicious network connections, or quarantining affected files. Automated response systems have been shown to significantly improve incident response efficiency by reducing detection-to-response time and minimizing human intervention [3], [45]. This capability is particularly beneficial in SME environments, where dedicated security teams may be limited. At the same time, the system maintains a human-in-the-loop approach, allowing users to review and validate alerts through an intuitive interface, ensuring transparency and control over critical security decisions.

A critical aspect of the proposed methodology is the implementation of a secure and tamper-proof logging mechanism to ensure the integrity and reliability of recorded security events. Traditional logging systems are vulnerable to manipulation, making them unreliable for forensic analysis. To address this issue, the system incorporates cryptographic hashing techniques, specifically SHA-256, to generate unique identifiers for each log entry. These logs are further secured using advanced cryptographic logging mechanisms such as forward-secure logging and hash chaining, which prevent unauthorized modification of stored data [35], [36]. In addition, blockchain-based logging approaches are utilized to provide immutability and non-repudiation,

ensuring that once log data is recorded, it cannot be altered or deleted without detection [38], [39], [41]. The use of cryptographic structures such as Merkle trees further enhances the efficiency and integrity verification of stored logs [40]. This approach ensures that forensic evidence remains trustworthy and supports compliance with regulatory requirements related to data integrity and auditing.

Furthermore, the methodology incorporates resilience and fault-tolerance mechanisms to ensure continuous system operation under adverse conditions. The endpoint agent is equipped with a watchdog-based self-recovery mechanism that automatically restarts monitoring processes in the event of failure or termination, ensuring persistent system visibility. Such self-adaptive system behavior is aligned with modern software engineering practices that emphasize feedback loops and system resilience [49]. Additionally, the system supports offline operation by temporarily storing logs locally when network connectivity is unavailable and synchronizing them with the central system once connectivity is restored. This capability ensures that critical security data is preserved even during network disruptions, enhancing the reliability and robustness of the system.

Overall, the methodology of the ForLENS system emphasizes integration, efficiency, and practicality by combining multiple advanced cybersecurity techniques into a unified framework. By bridging the gap between lightweight endpoint monitoring, intelligent anomaly detection, threat intelligence integration, secure logging, and automated response, the proposed approach provides a scalable and cost-effective solution tailored to the unique requirements of SME environments. The modular design of the system also allows for future enhancements, including the integration of more advanced machine learning models, expanded threat intelligence sources, and improved automation capabilities. This adaptability ensures that the system can evolve in response to emerging threats and technological advancements, thereby maintaining long-term effectiveness in dynamic cybersecurity landscapes.

2.2. System Architecture / Framework Design

The ForLENS system is designed using a modular, layered architecture that enables seamless integration of multiple cybersecurity functionalities while maintaining scalability, flexibility, and efficiency suitable for Small and Medium Enterprise (SME) environments. The architecture follows a distributed endpoint-centric approach combined with a centralized analytical and control layer, ensuring that data collection, threat detection, response, and forensic logging operate as a unified system. At a high level, the system consists of four primary components: the lightweight endpoint monitoring agent with embedded deception capabilities, the machine learning-based anomaly detection and threat intelligence engine, the secure blockchain-based logging module, and the SOAR-inspired alert management and response interface. These components are interconnected through a structured data flow pipeline that ensures continuous monitoring, intelligent analysis, automated response, and secure evidence preservation.

At the foundation of the architecture lies the endpoint layer, where the lightweight monitoring agent is deployed across individual systems. This agent serves as the primary data collection point and operates continuously to capture system-level telemetry, including process execution details, file system changes, network connections, and user activity patterns. The agent is designed with minimal resource consumption to ensure compatibility with SME infrastructure while maintaining high-frequency monitoring. In addition to passive data collection, the endpoint layer incorporates an embedded Server Message Block (SMB)-based honeypot module, which introduces deception capabilities directly within the host environment. This module simulates vulnerable shared resources to attract malicious actors, enabling early detection of unauthorized access attempts, lateral movement, and insider threat behavior. By integrating deception at the endpoint level, the system enhances visibility into attacker actions that would otherwise bypass traditional monitoring mechanisms.

The collected telemetry data is transmitted to the analytical layer, which forms the core intelligence component of the system. This layer integrates machine learning models with external threat intelligence sources to perform comprehensive threat analysis. The anomaly detection engine processes incoming data to establish behavioral baselines and identify deviations that may indicate malicious activity. The use of unsupervised and semi-supervised learning techniques allows the system to detect both known and unknown threats without relying

solely on predefined signatures. In parallel, the threat intelligence integration module enriches the analyzed data by correlating it with external sources, such as known malicious IP addresses, domains, and file hashes obtained from open-source intelligence platforms. This dual-layer analysis improves detection accuracy and reduces false positives by combining behavioral insights with contextual threat information. Furthermore, explainable artificial intelligence techniques are incorporated to provide interpretable outputs, allowing users to understand the reasoning behind detected anomalies and improving trust in the system.

The architecture also includes a secure logging layer responsible for ensuring the integrity and reliability of all generated security data. Instead of storing raw logs directly in a centralized database, the system employs a cryptographic approach where log entries are grouped, hashed using SHA-256, and anchored to a blockchain-based infrastructure. This design ensures that all recorded events are immutable and verifiable, preventing unauthorized modifications or deletions. By storing only the hash values on the blockchain while optionally maintaining encrypted logs in distributed storage systems, the architecture achieves both data integrity and storage efficiency. The secure logging layer plays a critical role in supporting forensic investigations, as it provides a trustworthy record of system activities that can be independently verified at any time.

At the top of the architecture is the presentation and control layer, which is implemented through a SOAR-inspired alert interface. This component acts as the central management system, aggregating alerts generated by the analytical layer and presenting them through a user-friendly dashboard. The interface is designed to simplify complex security information, enabling non-expert users to understand threats through clear visualizations and contextual explanations. In addition to visualization, this layer facilitates automated response actions based on predefined rules and intelligent decision-making processes. Actions such as isolating endpoints, terminating suspicious processes, blocking network connections, and quarantining files can be executed automatically or with user approval, depending on system configuration. This integration of detection and response capabilities significantly reduces the time required to mitigate threats and enhances overall system effectiveness.

The overall architecture emphasizes seamless interaction between components through a continuous data flow mechanism. The endpoint agent generates telemetry, which is analyzed by the machine learning and threat intelligence engine, producing alerts that are both actionable and contextually enriched. These alerts are then presented to the SOAR interface for visualization and response, while all related events are simultaneously secured within the blockchain-based logging system to ensure forensic integrity. The modular nature of the architecture allows each component to function independently while contributing to the overall system, enabling scalability and ease of future enhancements. By combining distributed monitoring, centralized intelligence, secure logging, and automated response within a single cohesive framework, the ForLENS architecture effectively addresses the limitations of existing solutions and provides a robust cybersecurity model tailored for SME environments.

2.3. Description of Components (All Four Modules)

The ForLENS system is composed of four tightly integrated components, each designed to perform a specific function within the overall cybersecurity framework while collectively contributing to a unified detection, response, and forensic analysis pipeline. These components include the lightweight endpoint monitoring agent with embedded deception capabilities, the machine learning-based anomaly detection and threat intelligence engine, the blockchain-based secure logging module, and the SOAR-inspired security alert interface. Each component is implemented with a focus on efficiency, scalability, and practicality to ensure suitability for SME environments.

The first component, the lightweight endpoint monitoring agent, serves as the foundation of the system by providing continuous visibility into endpoint-level activities. This agent is deployed directly on host systems and is responsible for collecting real-time telemetry data, including process creation and termination events, file system modifications such as creation, deletion, and updates, network connections, and user interaction patterns. The agent is implemented using a resource-efficient design to ensure minimal CPU and memory usage, allowing it to operate effectively in environments with limited computational capacity. In addition to passive monitoring, the agent incorporates an embedded Server Message Block (SMB)-based honeypot, which introduces deception capabilities directly within the endpoint. This honeypot simulates vulnerable shared resources to attract malicious actors, enabling the detection of unauthorized

access attempts, lateral movement, and insider threats. When such interactions occur, the agent captures relevant details such as source IP addresses, timestamps, and accessed resources, providing valuable insights into attacker behavior. Furthermore, the agent includes automated containment mechanisms that allow it to take immediate action upon detecting suspicious activity, such as quarantining malicious files or terminating suspicious processes. To ensure resilience against tampering, the agent is equipped with a watchdog-based self-recovery mechanism that automatically restarts the monitoring process if it is terminated, thereby maintaining continuous visibility even under adversarial conditions. Additionally, the agent supports offline operation by storing logs locally when network connectivity is unavailable and synchronizing them once connectivity is restored.

The second component, the machine learning-based anomaly detection and threat intelligence engine, acts as the analytical core of the system by processing the telemetry data generated by the endpoint agent. This component utilizes a combination of unsupervised and semi-supervised machine learning models to establish behavioral baselines and identify deviations that may indicate malicious activity. Techniques such as Isolation Forests, Autoencoders, and sequence-based models are employed to analyze patterns in system behavior, including login activities, file access frequencies, and process execution sequences. Unlike traditional signature-based systems, this approach enables the detection of previously unknown or evolving threats, making it particularly effective against insider attacks and zero-day exploits. To enhance detection accuracy and contextual awareness, the engine integrates threat intelligence data obtained from open-source platforms such as Malware Information Sharing Platform (MISP), Open Threat Exchange (OTX), and AbuseIPDB. This integration allows the system to correlate detected anomalies with known malicious indicators, such as suspicious IP addresses, domains, and file hashes, thereby improving confidence in generated alerts. Additionally, the component incorporates explainable artificial intelligence techniques to provide interpretable insights into detected anomalies, allowing users to understand the reasoning behind alerts and improving system transparency. The output of this component consists of enriched and prioritized alerts that are forwarded to the response and visualization layer.

The third component, the blockchain-based secure logging module, is responsible for ensuring the integrity, immutability, and reliability of all recorded security events. This component

addresses the critical challenge of log tampering by employing cryptographic hashing and blockchain technology to create a verifiable record of system activities. Log entries generated by the endpoint agent and analytical components are grouped into batches and processed using a cryptographic hash function, specifically SHA-256, to generate unique hash values. These hash values are then anchored to a blockchain network through smart contracts, ensuring that once recorded, they cannot be altered or deleted. By storing only the hash values on the blockchain, the system achieves storage efficiency while maintaining strong integrity guarantees. In addition to blockchain anchoring, the system may utilize distributed storage solutions such as InterPlanetary File System (IPFS) or searchable databases like Elasticsearch to store encrypted log data for retrieval and analysis. During forensic investigations, the integrity of logs can be verified by recalculating the hash values and comparing them with the corresponding blockchain entries, enabling the detection of any unauthorized modifications. This component plays a crucial role in supporting incident response and compliance requirements by providing a trustworthy and tamper-proof audit trail.

The fourth component, the SOAR-inspired security alert interface, functions as the central control and visualization layer of the system. This component aggregates alerts generated by the anomaly detection engine and presents them through a user-friendly dashboard designed for both technical and non-technical users. The interface performs alert normalization by converting data from different sources into a standardized format, enabling consistent analysis and presentation. It also incorporates prioritization mechanisms that rank alerts based on severity, potential impact, and contextual intelligence, allowing users to focus on the most critical threats. In addition to visualization, the interface enables automated response actions based on predefined rules and intelligent decision-making processes. These actions include isolating compromised endpoints, blocking malicious IP addresses, terminating suspicious processes, and quarantining affected files. The system also supports a human-in-the-loop approach, allowing users to review and approve actions when necessary, ensuring a balance between automation and control. Furthermore, the interface integrates explainable AI outputs to provide clear and understandable descriptions of detected threats, improving usability and reducing the cognitive burden on users. The adaptive learning capability of the interface allows it to refine its behavior based on user feedback, further enhancing its effectiveness over time.

Together, these four components form a cohesive and integrated cybersecurity framework that addresses the key challenges faced by SMEs. The endpoint agent provides real-time data collection and early detection capabilities, the machine learning and threat intelligence engine performs intelligent analysis, the secure logging module ensures the integrity of recorded events, and the SOAR interface enables effective visualization and response. This modular yet interconnected design allows the system to deliver comprehensive security functionality while maintaining efficiency and scalability, making it a practical solution for resource-constrained environments.

2.4. Data Flow & Integration

The effectiveness of the ForLENS framework depends on a well-defined, continuous data flow that seamlessly connects all four components into a unified detection–analysis–response–forensics pipeline. The system adopts an event-driven architecture in which security-relevant activities generated at the endpoint are transformed into structured events, enriched with contextual intelligence, acted upon in real time, and finally preserved as tamper-evident records for audit and forensic purposes. This end-to-end flow ensures that no stage of the security lifecycle operates in isolation and that information produced at one stage directly informs subsequent decisions across the system.

The data flow begins at the endpoint layer, where the lightweight monitoring agent continuously captures system telemetry from multiple sources, including process execution (e.g., PID, executable path, parent process, command line), file system operations (create, modify, delete, and hash values), network connections (local/remote addresses, ports, protocols), and user activity patterns. In parallel, the embedded SMB-based honeypot generates high-fidelity signals when deceptive resources are accessed, particularly capturing events such as Windows Security Event ID 5140, which indicates access to shared resources. All captured observations are normalized into a consistent event schema to support downstream processing. Each event includes standardized fields such as timestamp, host identifier, user context, event type, and event-specific attributes. This normalization step is essential to ensure interoperability between heterogeneous data sources and to enable efficient parsing by the analytical engine.

Once normalized, events are forwarded through a secure communication channel to the analytical layer. The transmission mechanism utilizes HTTPS-based communication to ensure confidentiality and integrity during transit. To maintain resilience in environments with intermittent connectivity, the agent employs a local buffering mechanism, where events are temporarily stored in structured JSON logs and transmitted in batches when connectivity is restored. This guarantees that no security-relevant data is lost due to network disruptions. Additionally, the agent implements back-pressure handling and rate control to prevent network congestion and to maintain stable system performance under high event volumes.

At the analytical layer, incoming events are ingested by the machine learning and threat intelligence engine through a streaming or micro-batch processing pipeline. The ingestion module performs validation, deduplication, and feature extraction, transforming raw events into feature vectors suitable for anomaly detection models. Behavioral baselines are maintained for hosts and users, allowing the system to detect deviations such as unusual login times, abnormal file access frequencies, or anomalous process chains. Unsupervised models (e.g., Isolation Forests) and sequence-aware models (e.g., LSTM where applicable) evaluate these features to assign anomaly scores. In parallel, a threat intelligence enrichment process queries external feeds (e.g., MISP, OTX, AbuseIPDB, URLHaus) to annotate events with reputation data, such as whether an IP address or domain is known to be malicious. A correlation engine then fuses anomaly scores with intelligence indicators to produce enriched alerts with higher confidence and reduced false positives. Explainability modules generate human-readable justifications for each alert, highlighting the features that contributed to the decision.

The enriched alerts are then propagated to the presentation and control layer, implemented as the SOAR-inspired interface. This layer performs alert normalization and prioritization, ranking alerts based on severity, confidence, and potential business impact. The interface presents alerts through a dashboard that provides contextual information, including affected assets, attack vectors, and recommended actions. Based on predefined playbooks and decision rules, the system can initiate automated response actions such as isolating the endpoint, terminating malicious processes, blocking suspicious network connections, or quarantining files. The integration supports both fully automated responses and a human-in-the-loop mode, where users can approve or override actions. Feedback from user decisions is captured and fed back into the

system to refine prioritization and reduce future false positives, establishing an adaptive learning loop.

Concurrently with detection and response, all events and alert metadata are routed to the secure logging module to ensure forensic integrity. Instead of storing logs in a mutable centralized repository, the system groups events into batches and computes cryptographic hashes using SHA-256. These hashes are then anchored to a blockchain network via smart contracts, creating an immutable reference for each batch. The original logs may be stored in encrypted form in auxiliary storage (e.g., IPFS for distributed storage or Elasticsearch for indexed search), while the blockchain maintains the tamper-evident proofs. During verification, the system recomputes hashes from stored logs and compares them against the blockchain entries to detect any discrepancies. This dual-storage strategy achieves both efficient retrieval and strong integrity guarantees without incurring excessive storage overhead on-chain.

Integration across components is maintained through clearly defined interfaces and data contracts. The endpoint agent outputs standardized event records; the analytical engine consumes these records and produces enriched alerts; the SOAR interface consumes alerts and triggers response actions; and the logging module receives both raw events and alert metadata for integrity anchoring. Time synchronization across components is enforced to maintain consistent event ordering, and unique identifiers are assigned to events to support traceability across the pipeline. Error handling and retry mechanisms are implemented at each interface to ensure reliability, while modular boundaries allow components to be updated or scaled independently without disrupting the overall system.

Overall, the ForLENS data flow and integration strategy ensures a continuous, resilient, and verifiable security lifecycle, where endpoint telemetry is transformed into actionable intelligence, responses are executed in near real time, and all activities are preserved with cryptographic assurance for future audit and forensic analysis.

2.5. Tools and Technologies Used

The implementation of the ForLENS framework relies on a carefully selected set of tools and technologies that collectively support real-time monitoring, intelligent analysis, secure

communication, automated response, and tamper-proof logging, while maintaining the lightweight and resource-efficient requirements of Small and Medium Enterprise (SME) environments. The choice of technologies is driven by factors such as performance efficiency, ease of integration, scalability, and compatibility with the modular architecture of the system.

At the core of the endpoint layer, the monitoring agent is implemented using the Python programming language due to its simplicity, flexibility, and extensive support for system-level operations and rapid development. Python enables efficient interaction with operating system resources while allowing seamless integration of various libraries required for monitoring, data processing, and communication. The agent utilizes the *psutil* library to capture real-time system information, including process details, CPU usage, memory consumption, and network connections. File system monitoring is achieved through Python's built-in modules and event-based tracking mechanisms, enabling the detection of file creation, modification, and deletion activities. For network monitoring and connection tracking, the agent leverages system-level APIs exposed through Python libraries to capture both local and remote connection details. Additionally, Windows Event Logs are accessed to detect SMB-based honeypot interactions, particularly by monitoring specific event identifiers such as Event ID 5140, which indicates access to shared resources.

To ensure secure and efficient data handling, the agent employs the *json* module for structured logging, enabling consistent event formatting and easy transmission across components. Cryptographic operations, including hashing for log integrity, are implemented using the *hashlib* library, specifically utilizing the SHA-256 algorithm to generate secure hash values for log entries. Multithreading is handled using Python's *threading* module, allowing the agent to perform monitoring, logging, and communication tasks concurrently without affecting system performance. For communication between the endpoint agent and the central analytical system, the *requests* library is used to establish secure HTTP (HTTPS) connections, ensuring encrypted data transmission. This communication mechanism also supports retry logic and buffering to handle intermittent network connectivity, enabling reliable data transfer even in unstable environments.

The analytical layer incorporates machine learning technologies for anomaly detection and behavioral analysis. Depending on the complexity of the implementation, frameworks such as *Scikit-learn* are used to implement models like Isolation Forests for anomaly detection due to their efficiency and suitability for unsupervised learning. For more advanced sequence-based analysis, libraries such as *TensorFlow* or *Keras* may be utilized to implement Long Short-Term Memory (LSTM) networks, enabling the system to capture temporal patterns in user and system behavior. Feature extraction and preprocessing are handled using data manipulation libraries such as *NumPy* and *Pandas*, which facilitate efficient handling of large volumes of telemetry data. The integration of threat intelligence is achieved through APIs provided by external platforms such as MISP, Open Threat Exchange (OTX), and AbuseIPDB, enabling real-time retrieval of information related to malicious IP addresses, domains, and file hashes. These APIs are accessed through HTTP requests and parsed into the system to enrich detected anomalies with contextual information.

The secure logging component leverages cryptographic and distributed ledger technologies to ensure data integrity and immutability. The SHA-256 hashing algorithm is used to generate unique identifiers for log batches, while blockchain integration is achieved using platforms such as Ethereum, particularly in a Proof-of-Authority (PoA) configuration for improved efficiency and reduced computational overhead. Smart contracts are deployed to store hash values on the blockchain, ensuring that log integrity can be verified at any time. For off-chain storage of logs, distributed file systems such as the InterPlanetary File System (IPFS) may be used to provide secure and decentralized storage, while tools like Elasticsearch can be integrated to enable efficient searching and indexing of log data. This hybrid storage approach ensures both scalability and integrity without excessive resource consumption.

The presentation and control layer is implemented using web-based technologies to provide a user-friendly interface for monitoring and response. Frameworks such as *Flask* or *Django* may be used for backend development, enabling the creation of RESTful APIs for communication between system components. The frontend interface can be developed using standard web technologies such as HTML, CSS, and JavaScript, along with visualization libraries like Chart.js or D3.js to present alerts and system metrics in an intuitive manner. The SOAR functionality is implemented through rule-based automation scripts that define response actions based on

specific conditions, allowing the system to automatically execute tasks such as isolating endpoints, blocking network connections, or terminating processes. These actions are triggered through API calls to the endpoint agent or system-level commands, ensuring rapid response to detected threats.

To support deployment and scalability, the system may utilize virtualization or containerization technologies such as Docker, enabling consistent environments across different systems. Version control is maintained using Git, ensuring proper tracking of code changes and collaboration among team members. Additionally, development and testing environments may include tools such as virtual machines or sandboxed environments to simulate attack scenarios and evaluate system performance under controlled conditions.

Overall, the combination of these tools and technologies enables the ForLENS system to achieve a balance between functionality, performance, and practicality. By leveraging lightweight programming frameworks, efficient machine learning libraries, secure communication protocols, and robust cryptographic techniques, the system successfully implements an integrated cybersecurity solution that is both technically advanced and suitable for real-world SME deployment.

2.6. Commercialization Aspect of the Product

The ForLENS framework is designed not only as a research prototype but also as a commercially viable cybersecurity solution tailored specifically for Small and Medium Enterprises (SMEs), addressing a critical gap in the current cybersecurity market. SMEs constitute a substantial portion of global economic activity, yet they remain significantly underserved in terms of affordable and effective cybersecurity solutions. Studies and industry reports highlight that SMEs face increasing cyber threats while lacking the financial and technical capacity to implement advanced security infrastructures [9], [15], [31]. Most enterprise-grade tools, such as Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Security Orchestration, Automation, and Response (SOAR) platforms, are typically priced at premium levels and require specialized expertise for deployment and management, making them inaccessible to organizations with limited budgets and technical resources [20], [21], [48]. In this context, ForLENS introduces a cost-effective, lightweight, and integrated cybersecurity solution

that can serve as a practical alternative for SMEs seeking enhanced protection without incurring high operational costs.

From a product deployment perspective, ForLENS can be implemented using a Software-as-a-Service (SaaS) or hybrid architecture, enabling flexible adoption across different organizational environments. In this model, lightweight endpoint agents are deployed across user systems, while centralized analytics and management functionalities are hosted either in the cloud or on-premises, depending on organizational requirements. Such deployment flexibility aligns with modern cybersecurity trends, where hybrid and cloud-based solutions are increasingly adopted to balance scalability, performance, and data control [20], [21]. The modular design of ForLENS further enhances its commercial potential by allowing individual components—such as endpoint monitoring, anomaly detection, secure logging, and automated response—to be offered either as standalone services or as integrated packages. This modularity enables organizations to select features based on their specific security needs and financial constraints, thereby increasing accessibility and market adaptability.

A key competitive advantage of ForLENS lies in its ability to integrate multiple advanced cybersecurity capabilities into a single unified platform. Traditional security approaches often require organizations to deploy separate tools for monitoring, detection, logging, and response, leading to increased complexity and higher costs. In contrast, ForLENS consolidates these functionalities into a cohesive framework, simplifying security operations while enhancing efficiency. The incorporation of deception mechanisms, such as embedded honeypots, provides an additional proactive layer of defense by detecting attacker behavior early in the attack lifecycle [1], [2], [8]. Furthermore, the integration of machine learning techniques enables intelligent anomaly detection, allowing the system to identify previously unknown threats based on behavioral deviations [22], [25]. The inclusion of threat intelligence platforms enhances detection accuracy by correlating observed activities with known malicious indicators, thereby improving situational awareness [27], [28]. Additionally, the implementation of blockchain-based secure logging introduces a unique value proposition by ensuring tamper-proof forensic records, which are essential for compliance, auditing, and incident investigation [38], [39], [41].

From a market positioning perspective, ForLENS is targeted at SMEs operating across diverse sectors such as finance, healthcare, education, and retail, where data protection and regulatory compliance are critical. These sectors often require reliable security mechanisms but lack access to affordable enterprise-level solutions. The system can be commercialized using a subscription-based pricing model with tiered service offerings. For instance, a basic tier may include essential features such as endpoint monitoring and anomaly detection, while advanced tiers may provide full integration of threat intelligence, blockchain-based logging, and automated SOAR capabilities. Such tiered pricing strategies are widely adopted in modern SaaS business models to ensure scalability and accessibility while enabling revenue growth through feature upgrades [48]. This approach allows SMEs to adopt cybersecurity solutions progressively based on their evolving needs and budgets.

Ease of deployment and usability are also critical factors influencing the commercialization success of cybersecurity products. The lightweight design of the ForLENS endpoint agent ensures minimal system requirements, enabling rapid deployment across multiple devices without significant infrastructure changes. This is particularly important for SMEs, where complex deployment processes can hinder adoption. Moreover, the system incorporates a user-friendly dashboard with explainable alerting mechanisms, allowing users with limited technical expertise to understand and respond to security events effectively. Explainability in cybersecurity systems is increasingly recognized as an essential feature, as it enhances user trust and facilitates informed decision-making [26], [30]. Additionally, the automated response capabilities of the system reduce reliance on manual intervention, thereby lowering operational overhead and improving response efficiency, which is a key advantage in resource-constrained environments [3], [45].

From a long-term business sustainability perspective, ForLENS can be continuously enhanced through periodic updates that incorporate new threat intelligence feeds, improved machine learning models, and expanded automation capabilities. The integration of external threat intelligence sources ensures that the system remains up-to-date with evolving threat landscapes, while advancements in machine learning can improve detection accuracy over time. Furthermore, strategic partnerships with cybersecurity vendors and integration with existing enterprise tools can extend the system's functionality and market reach. The ability to support

compliance requirements through secure logging and audit trails also increases its attractiveness for organizations that must adhere to data protection regulations such as GDPR and HIPAA [42], [43].

Overall, the commercialization potential of the ForLENS framework is significant, as it addresses a clearly identified gap in the cybersecurity market by providing a lightweight, integrated, and cost-effective solution tailored specifically for SMEs. By combining advanced technologies such as machine learning, threat intelligence, deception mechanisms, and blockchain-based logging with practical deployment and pricing strategies, the system bridges the gap between enterprise-level security capabilities and the accessibility requirements of smaller organizations. This positions ForLENS as a viable and scalable product for real-world adoption, capable of enhancing the cybersecurity resilience of SMEs while maintaining affordability and ease of use.

2.7. Testing and Implementation

The implementation of the ForLENS framework was carried out using a modular development approach, where each component was developed, tested, and integrated progressively to ensure system stability and functional correctness. The system was deployed in a controlled experimental environment designed to simulate real-world Small and Medium Enterprise (SME) conditions, consisting of endpoint machines running the lightweight monitoring agent and a centralized system responsible for analysis, visualization, and logging. The primary objective of the implementation phase was to validate the effectiveness of the proposed architecture in detecting, analyzing, responding to, and securely recording cybersecurity events under various attack scenarios while maintaining low resource consumption.

The endpoint monitoring agent was implemented using Python and deployed on test machines to continuously collect system telemetry. The agent was configured to monitor process execution, file system activities, network connections, and user interactions in real time. The embedded SMB-based honeypot was set up within the endpoint environment to simulate a vulnerable shared resource, allowing the system to detect unauthorized access attempts and lateral

movement activities. The agent was further enhanced with automated containment capabilities, enabling it to respond immediately to suspicious activities by quarantining files or terminating processes. To ensure robustness, the agent was tested under different operational conditions, including normal usage, high system load, and intentional termination attempts. The watchdog mechanism successfully restored the agent in cases where it was manually or programmatically terminated, demonstrating its resilience against tampering. Additionally, offline functionality was tested by disconnecting the network, during which the agent continued to log events locally and successfully synchronized the data once connectivity was restored.

The analytical component of the system was implemented to process incoming telemetry data and perform anomaly detection using machine learning techniques. Test datasets were generated through simulated user behavior and attack scenarios, including abnormal login times, excessive file access, unauthorized process execution, and suspicious network activity. The anomaly detection models were evaluated based on their ability to correctly identify deviations from normal behavior, and the integration of threat intelligence feeds was tested by introducing known malicious indicators such as blacklisted IP addresses and domains. The system successfully enriched detected anomalies with contextual information, improving the accuracy and reliability of alerts. Furthermore, explainability mechanisms were evaluated by verifying that the system could provide meaningful descriptions of detected threats, enabling users to understand the underlying causes of alerts.

The secure logging module was implemented using cryptographic hashing and blockchain integration to ensure data integrity and immutability. During testing, log entries generated by the system were grouped and hashed using the SHA-256 algorithm, and the resulting hash values were recorded on a blockchain network through smart contracts. To evaluate the effectiveness of the tamper detection mechanism, controlled modifications were introduced to stored log data. The system successfully detected all instances of tampering by identifying mismatches between recalculated hashes and the corresponding blockchain entries, confirming the reliability of the logging mechanism. Performance testing also demonstrated that the storage overhead was minimal, as only hash values were stored on the blockchain, while full logs were maintained in auxiliary storage.

The SOAR-inspired alert interface was implemented as a web-based dashboard to provide real-time visualization and management of security events. The interface was tested for usability by displaying alerts generated from various attack scenarios and evaluating the clarity of information presented to users. The alert prioritization mechanism effectively ranked threats based on severity and contextual relevance, allowing users to focus on critical incidents. Automated response actions were tested by triggering predefined rules in response to detected threats, such as isolating endpoints, blocking network connections, and quarantining files. These actions were executed successfully with minimal delay, demonstrating the system's ability to reduce response time and mitigate threats efficiently. The human-in-the-loop functionality was also validated, allowing users to review and approve actions when necessary, ensuring control and transparency in decision-making.

To comprehensively evaluate the system, multiple test scenarios were designed to simulate real-world attack conditions. These included unauthorized SMB access to the honeypot, execution of malicious files, abnormal user behavior patterns, and network-based attacks. The system consistently demonstrated the ability to detect threats in real time, generate meaningful alerts, and execute appropriate response actions. Performance metrics such as detection time, response time, CPU usage, and memory consumption were monitored throughout the testing phase. The results indicated that the system maintained low resource usage, with minimal impact on endpoint performance, while achieving rapid detection and response capabilities.

Overall, the implementation and testing phase confirmed that the ForLENS framework operates effectively as an integrated cybersecurity solution. The system successfully combines lightweight endpoint monitoring, intelligent anomaly detection, secure logging, and automated response within a single cohesive architecture. The experimental results validate the practicality of the proposed approach and demonstrate its potential for real-world deployment in SME environments, where efficiency, cost-effectiveness, and ease of use are critical factors.

3. RESULTS AND DISCUSSION

3.1. Results

The implementation of the ForLENS framework was evaluated using multiple simulated attack scenarios designed to reflect real-world cybersecurity threats in Small and Medium Enterprise (SME) environments. The system demonstrated effective real-time monitoring and detection capabilities across all components. The endpoint monitoring agent successfully captured system-level activities, including process execution, file modifications, network connections, and honeypot interactions. The embedded SMB-based honeypot effectively detected unauthorized access attempts and lateral movement by capturing relevant event data such as source IP addresses and access timestamps.

The machine learning-based anomaly detection engine was able to identify deviations from normal behavior patterns, including abnormal login times, excessive file access, and suspicious process execution. The integration of threat intelligence further enhanced detection accuracy by correlating anomalies with known malicious indicators. The system generated alerts in real time, which were successfully processed and displayed through the SOAR-inspired interface. Automated response actions, such as process termination, endpoint isolation, and file quarantine, were executed with minimal delay, demonstrating the system's ability to mitigate threats efficiently.

The secure logging module ensured the integrity of all recorded events by generating cryptographic hashes and anchoring them to the blockchain. During testing, any attempt to modify log data was successfully detected, confirming the reliability of the tamper-proof logging mechanism. Overall, the system maintained low resource consumption while delivering fast detection and response capabilities, making it suitable for SME environments.

Test ID	Objective	Expected Outcome	Pass Criteria
TC-01	Verify activity monitoring	Capture process/file/network events	>95% events captured and forwarded

TC-02	Validate honeypot trigger	SMB scan detection and alert	Latency < 1 second
TC-03	Test containment speed	Automatic file quarantine	Response within < 2 seconds
TC-04	Log Storage Efficiency	Anchoring 10k entries	Storage < 0.65 MB
TC-05	Agent Survivability	forced process termination	Auto-restart in < 2 seconds
TC-06	Resource Overhead	Monitor CPU/RAM impact	CPU 2–4%; RAM 60–90MB
TC-07	Offline Continuity	Network disruption test	0% data loss via local buffering

Table 2: Test Cases

3.2. Research Findings

The evaluation of the ForLENS framework revealed several key findings regarding its effectiveness and practicality. Firstly, the integration of multiple cybersecurity techniques into a single framework significantly improved overall system performance compared to standalone solutions. The combination of endpoint monitoring, machine learning-based detection, threat intelligence, secure logging, and automated response enabled a comprehensive approach to cybersecurity, addressing multiple stages of the attack lifecycle.

Secondly, the use of behavioral anomaly detection proved effective in identifying both known and unknown threats, particularly insider attacks and abnormal user activities that are difficult to detect using traditional signature-based methods. The incorporation of threat intelligence further enhanced detection accuracy by providing contextual information, reducing false positives, and improving confidence in alerts.

Additionally, the inclusion of a deception-based honeypot at the endpoint level provided valuable insights into attacker behavior, enabling early detection of unauthorized access attempts. The secure logging mechanism ensured that all security events were recorded in a tamper-proof manner, supporting reliable forensic analysis and auditability. Furthermore, the

automated response capabilities significantly reduced response time and minimized the need for manual intervention, making the system practical for environments with limited security expertise.

3.2.1. Deception Efficacy and Operational Resilience

The experimental results validate that embedding deception directly at the host level significantly reduces detection latency compared to network-level systems. The integrated agent maintained a minimal footprint (2–4% CPU), proving its suitability for SME hardware. Furthermore, the watchdog mechanism proved critical; the agent survived over 90% of manual termination attempts, maintaining a resilient "chain of trust" for forensic evidence collection.

3.2.2. Storage Efficiency vs. Verification Speed

LogStamping achieved superior storage efficiency, outperforming EngraveChain by 95% in storage requirements. While the grouped hash approach results in a verification time of ~155s for 10,000 entries, this trade-off is preferable for SMEs where storage is at a premium and

3.3. Discussion

The results demonstrate that the ForLENS framework successfully addresses the key limitations identified in existing cybersecurity solutions for SMEs. Unlike traditional systems that are resource-intensive and complex, the proposed framework provides a lightweight and integrated solution that balances performance, efficiency, and usability. The modular architecture allows each component to function independently while contributing to the overall system, ensuring scalability and flexibility.

The integration of machine learning and threat intelligence enhances detection capabilities, while the SOAR-inspired interface improves usability by providing clear and actionable insights. The blockchain-based logging mechanism adds a layer of trust and reliability, which is critical for forensic investigations and compliance. However, certain limitations remain, such as the reliance on simulated datasets for testing and the need for further optimization of machine learning models for large-scale deployment.

Despite these limitations, the framework demonstrates strong potential for real-world application, particularly in SME environments where cost, simplicity, and effectiveness are critical factors. Future improvements can focus on expanding threat intelligence sources, enhancing model accuracy, and conducting large-scale real-world evaluations to further validate the system's performance.

3.4. Summary of Each Student's Contribution

The development of the ForLENS framework was carried out collaboratively, with each team member responsible for a specific component of the system. The first component involved the design and implementation of the lightweight endpoint monitoring agent with embedded SMB-based deception capabilities, enabling real-time telemetry collection, early threat detection, and automated containment. The second component focused on the development of the machine learning-based anomaly detection and threat intelligence engine, responsible for analyzing collected data, identifying anomalies, and enriching alerts with contextual intelligence. The third component implemented the secure logging mechanism using cryptographic hashing and blockchain technology to ensure the integrity and immutability of security logs. The fourth component developed the SOAR-inspired alert interface, which provides real-time visualization, prioritization of alerts, and automated response actions. Together, these contributions form a cohesive and integrated cybersecurity framework that delivers comprehensive detection, response, and forensic capabilities tailored for SME environments.

4. Conclusion

The increasing frequency and sophistication of cyber threats have made cybersecurity a critical requirement for modern organizations, particularly for Small and Medium Enterprises (SMEs) that often operate with limited resources and lack access to advanced security infrastructures. This research addressed the challenge of developing an effective, lightweight, and integrated cybersecurity solution tailored specifically for SME environments. The proposed system, ForLENS, was designed to provide real-time threat detection, automated response, and secure forensic logging within a unified framework, overcoming the limitations of traditional standalone security tools.

The implementation of the ForLENS framework successfully demonstrated the feasibility of integrating multiple advanced cybersecurity techniques into a single cohesive system. The lightweight endpoint monitoring agent provided continuous visibility into system activities while maintaining minimal resource consumption, making it suitable for resource-constrained environments. The inclusion of an embedded deception mechanism enhanced the system's ability to detect unauthorized access attempts and attacker behavior at early stages. The machine learning-based anomaly detection engine, combined with threat intelligence integration, enabled the identification of both known and unknown threats, improving detection accuracy and reducing reliance on signature-based approaches. Furthermore, the secure logging component ensured the integrity and immutability of security data through cryptographic hashing and blockchain anchoring, providing reliable support for forensic investigations. The SOAR-inspired alert interface facilitated real-time visualization and automated response, significantly reducing the time required to mitigate threats and improving overall system usability.

The evaluation of the system through simulated attack scenarios confirmed its effectiveness in detecting abnormal behavior, responding to threats in real time, and maintaining secure and verifiable logs. The results demonstrated that the system achieves a balance between performance and functionality, delivering enterprise-level security capabilities in a lightweight and cost-effective manner. This highlights the practicality of the proposed solution for SMEs, where simplicity, efficiency, and affordability are essential considerations.

In conclusion, the ForLENS framework successfully addresses the identified research problem by providing an integrated, scalable, and efficient cybersecurity solution that combines endpoint monitoring, intelligent threat detection, secure logging, and automated response within a single architecture. The system contributes to the field of cybersecurity by demonstrating how advanced technologies can be adapted and optimized for SME environments, bridging the gap between high-end enterprise solutions and accessible security tools.

4.1. Future Work

While the ForLENS framework demonstrates strong potential, several areas can be further enhanced to improve its effectiveness and applicability in real-world environments. Future work may focus on incorporating more advanced machine learning models, including deep learning techniques, to improve detection accuracy and adaptability to evolving threat patterns. The integration of additional threat intelligence sources can further enhance contextual awareness and reduce false positives. Expanding the scope of deception mechanisms beyond SMB-based honeypots to include other protocols and services can improve detection coverage across different attack vectors.

Additionally, large-scale real-world deployment and testing across diverse SME environments would provide more comprehensive validation of the system's performance and scalability. Further improvements can also be made in optimizing system performance, particularly in handling high volumes of data and ensuring efficient resource utilization. Enhancing the user interface with more advanced visualization techniques and customizable dashboards can improve usability and decision-making for non-technical users. Finally, incorporating adaptive learning mechanisms that continuously update detection models based on new data and user feedback can further strengthen the system's long-term effectiveness.

5. References

- [1] S. Kandanaarachchi, H. Ochiai, and A. Rao, "Honeyboost: Boosting honeypot performance with data fusion and anomaly detection," arXiv preprint arXiv:2105.02526, 2021. [Online]. Available: <https://arxiv.org/abs/2105.02526>
- [2] A. Javadpour, F. Ja'fari, T. Taleb, M. Shojafar, and C. Benzaïd, "A comprehensive survey on cyber deception techniques to improve honeypot performance," *Computers & Security*, vol. 140, art. no. 103792, Mar. 2024, doi: 10.1016/j.cose.2024.103792.
- [3] H. Karlzén and T. Sommestad, "Automatic Incident Response Solutions: A Review of Proposed Solutions' Input and Output," in *Proc. 18th Int. Conf. Availability, Reliability and Security (ARES 2023)*, Benevento, Italy, Aug.–Sep. 2023, pp. 1–9, doi: 10.1145/3600160.3605066.
- [4] K. R. Ismail, Z. A. Brata, G. A. Nelistiani, S. Heo, H. Kim, and H. Kim, "Toward Robust Security Orchestration and Automated Response in Security Operations Centers with a Hyper-Automation Approach Using Agentic Artificial Intelligence," *Information*, vol. 16, no. 5, art. 365, 2025, doi: 10.3390/info16050365.
- [5] R. Holbel, J. Yerby, and W. Smith, "Utilizing Virtualized Honeypots for Threat Hunting, Malware Analysis, and Reporting," *Issues in Information Systems*, vol. 25, no. 1, pp. 265–278, 2024, doi: 10.48009/1_iis_2024_122
- [6] S. Lanz, S. L.-R. Pignol, P. Schmitt, H. Wang, M. Papaioannou, G. Choudhary, and N. Dragoni, "Optimizing Internet of Things Honeypots with Machine Learning: A Review," *Applied Sciences*, vol. 15, no. 10, art. 5251, 2025, doi: 10.3390/app15105251
- [7] S. T. Muntaha, F. Ashraf, I. Shahzad, and J. Iqbal, "Designing an Adaptive Honeypot for Advanced Cybersecurity Threat Detection," *Spectrum of Engineering Sciences*, vol. 3, no. 5, pp. 816–847, May 2025.
- [8] Z. Morić, V. Dakić, and D. Regvart, "Advancing Cybersecurity with Honeypots and Deception Strategies," *Informatics*, vol. 12, no. 1, art. 14, Jan. 2025, doi: 10.3390/informatics12010014
- [9] E. K. Isabirye, "Cloud Adoption and Digital Transformation Cybersecurity Consideration for SMEs," *Iconic Research and Engineering Journals*, vol. 8, no. 7, pp. 453, Jan. 2025.
- [10] Q. Wang, W. Ul Hassan, D. Li, K. Jee, X. Yu, K. Zou, J. Rhee, Z. Chen, W. Cheng, C. A. Gunter, and H. Chen, "You Are What You Do: Hunting Stealthy Malware via Data Provenance Analysis," in *Proc. 27th Network and Distributed System Security Symp. (NDSS)*, San Diego, CA, USA, Feb. 23–26, 2020, doi: 10.14722/ndss.2020.24167
- [11] P. Institute, "2024 Cost of Insider Threats Global Report," Ponemon Institute, 2024
- [12] G. & T. I. Russo, "Behavioral Analytics for Insider Threat Detection in Enterprise Environments," *ACE Journal*, 2025.

- [13] T. Michael, "Tolumichael, Best Open Source Threat Intelligence Platforms and Feeds," 2024. [Online]. Available: <https://tolumichael.com/best-open-source-threat-intelligence-platforms-and-feeds/>
- [14] SocRadar, "The Ultimate List of Free and Open-Source Threat Intelligence Feeds," 2025. [Online]. Available: <https://socradar.io/blog/the-ultimate-list-of-free-and-open-source-threat-intelligence-feeds/>
- [15] ENISA, "Cybersecurity for SMEs – Good practices and recommendations," 2024.
- [16] Verizon, "2024 Data Breach Investigations Report," 2024
- [17] IBM, "Cost of a Data Breach Report 2025," 2025.
- [18] M. G. F. S. M. Bishop., "Insider Threats in Cyber Security," 2010.
- [19] N. A. a. K. El-Khatib, On the effectiveness of machine learning in insider threat detection, vol. 5, IEEE Access, 2017.
- [20] A. Wagner, "Security Information and Event Management: A Review," 2021.
- [21] SentinelOne, "EDR vs. NDR vs. XDR: Choosing a Detection & Response Strategy," 2025.
- [22] F. T. Liu, "Isolation Forest," 2008.
- [23] I. Goodfellow, "Representation Learning with Autoencoders," 2016.
- [24] W. Du, "DeepLog: Anomaly Detection and Diagnosis from System Logs," ACM CCS, 2017.
- [25] R. C. a. S. Chawla, "Deep Learning for Anomaly Detection: A Survey," ACM Computing Surveys, 2019.
- [26] S. L. a. S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," NeurIPS, 2017.
- [27] CISA, "STIX Best Practices Guide v1.0.0," 2022
- [28] CIRCL.lu, "MISP User Guide," 2024.
- [29] OASIS, "TAXII 2.1 Specification," 2021.
- [30] M. R. e. al., "Why Should I Trust You? Explaining Predictions of Any Classifier," KDD, 2016
- [31] T. Anderson, "Emerging Cybersecurity Threats in Small Businesses: 2024 Insights," SecureTech Publications.
- [32] M. Silva, "SME Contributions and Cybersecurity Challenges in Sri Lanka," Regional Business Studies, 8(1), 40-58.

- [33] F. L. Greitzer and D. A. Frincke, "Combining traditional cyber security audit data with psychosocial data: Towards predictive modeling for insider threat mitigation," in *Insider Threats in Cyber Security*. Springer, 2010, pp. 85–113. doi: 10.1007/978-1-4419-7133-3_5.
- [34] M. B. Salem, S. Hershkop, and S. J. Stolfo, "A survey of insider attack detection research," in *Insider Attack and Cyber Security*. Springer, 2008, pp. 69–90. doi:10.1007/978-0-387-77322-3_5.
- [35] B. Schneier and J. Kelsey, "Secure audit logs to support computer forensics," *ACM Trans. Inf. Syst. Secur.*, vol. 2, no. 2, pp. 159–176, May 1999. doi: 10.1145/317087.317089.
- [36] M. Bellare and B. S. Yee, "Forward integrity for secure audit logs," University of California, San Diego, Tech. Rep., 1997. [Online]. Available: <https://cseweb.ucsd.edu/~bsy/pub/forward.pdf>
- [37] D. Ma and G. Tsudik, "A new approach to secure logging," *ACM Trans. Storage*, vol. 5, no. 1, pp. 1–21, Mar. 2009. doi: 10.1145/1502777.1502782.
- [38] I. Ahmad et al., "Towards blockchain-driven, secure and transparent audit logs," *arXiv preprint arXiv:1811.09944*, 2018. [Online]. Available: <https://arxiv.org/abs/1811.09944>
- [39] Y. Wang, Y. Liu, H. Zhang, and X. Li, "Blockchain-based forensic logging for tamper-proof evidence management," *Future Gener. Comput. Syst.*, vol. 110, pp. 756–766, Sep. 2020. doi: 10.1016/j.future.2019.10.024.
- [40] R. C. Merkle, "A digital signature based on a conventional encryption function," in *Advances in Cryptology — CRYPTO '87*, C. Pomerance, Ed. Springer Berlin Heidelberg, 1988, pp. 369–378. doi: 10.1007/3-540-48184-2_32.
- [41] L. Chen, Y. Xu, Z. Yang, and Y. Zhou, "Lightweight blockchain-based secure logging for IoT," *J. Netw. Comput. Appl.*, vol. 189, p. 103113, Sep. 2021. doi:10.1016/j.jnca.2021.103113
- [42] "GDPR: General Data Protection Regulation," Official Journal of the European Union, 2016. [Online]. Available: <https://gdpr-info.eu/>
- [43] "Health Insurance Portability and Accountability Act of 1996 (HIPAA)," CDC, 1996. [Online]. Available: <https://www.cdc.gov/phlp/publications/topic/hipaa.html>
- [44] P. Maniatis and M. Baker, "Secure history preservation through timeline entanglement," in *Proc. 11th USENIX Security Symposium*, San Francisco, CA, 2002, pp. 1–13.
- [45] S. R. Pulyala, A. G. Desetty, and V. D. Jangampet, "The Impact of Security Orchestration, Automation, and Response (SOAR) on Security Operations Center (SOC) Efficiency: A Comprehensive Analysis," **Turkish Journal of Computer and Mathematics Education**, vol. 10, no. 3, pp. 1545–1549, 2019. [Online]. Available: <https://turcomat.org/index.php/turkbilmat/article/view/14323/10355>
- [46] A. Horneman, J. Ray. "Benefits and Challenges of SOAR Platforms," March 2021. [Online].

Available: <https://www.sei.cmu.edu/blog/benefits-and-challenges-of-soar-platforms/>

[47] L. A. Odozor, L. Omini, S. N. Berko, U. Precious, Y. Nitzan, and K. Idowu, "An Incident Response Playbook Guide for Small and Medium Enterprises (SMEs)," *International Journal of Scientific and Management Research*, vol. 8, no. 7, pp. 145-157, July 2025. doi: 10.37502/IJSMR.2025.8712.

[48] H. Aver, "Commercial vs. open-source SIEM: pros and cons," Kaspersky Official Blog.

[Online]. Available: <https://www.kaspersky.com/blog/open-source-siem-hidden-costs/53589/>

[49] Y. Brun, G. Di Marzo Serugendo, C. Gacek, H. Giese, H. Kienle, M. Litoiu, H. Müller, M. Pezzè, and M. Shaw, "Engineering Self-Adaptive Systems through Feedback Loops," in *Proc. Int. Workshop on Software Engineering for Self-Adaptive Systems, LNCS 5525*, Springer, pp. 48–70, 2009. doi: 10.1007/978-3-642-02161-9_3

[50] P. F. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, "Deep reinforcement learning from human preferences," *arXiv preprint arXiv:1706.03741v4*, Feb. 2023. [Online]. Available: <https://arxiv.org/abs/1706.03741v4>